Introduction to Ansible (Part 2): Homework

Tasks

1. Create an inventory containing 2-3 Linux VMs (you can choose to create them with Hyper-V, Virtualbox, or host the machines on AWS/Azure/other cloud platform). The choice of OS is entirely up to you. You can use WSL or another VM as the Ansible master.
2. Using Ansible Galaxy, download a role that installs and configures Apache 2.x
   Try to understand how the role works, what tasks it uses and what variables you need to pass to it when including it in a playbook
3. Using the `ansible-galaxy role init` command, create a new role that will be used for creating UNIX users. This role should be able to do the following things:
   a. Create users from a given list
   b. Create groups for the users and assign each user to a list of groups
   c. Assign root privileges to specific users
   d. Add public SSH keys for the users as authorized keys
   e. Use Ansible Vault to encrypt the contents of the SSH public key files

Note: Use the concept of loops, conditions, variables, lists, and files in Ansible

4. Create a playbook file that will do the following things:
   a. Include the user management role, to create one user with root privileges (your personal user) and one non-privileged user (that will run the Apache service)
   b. Include the Apache installation role
   c. Include a task that will create an HTML file on the servers from a Jinja2 template
      Note: This HTML file can have any content you like. Make sure to use jinja2 variables, conditions, and loops (if possible) in the Jinja2 template for the HTML file
   d. Start the Apache service using the non-privileged user that you created
      Note: You might want to change the service port from 80 to another, non-privileged port (e.g., 8080), for security reasons (read more)
5. Test your playbook/role on the VMs you have created. Log into them to see if the results of the Ansible execution are as expected, and check if you can see the deployed HTML file in the browser.

Bonus

- Instead of passing host specific variables to the roles from the cli/playbook, or using the defaults/vars role directories, use host_vars and/or group_vars for the hosts in your inventory
- Add a functionality to the user management role for removing UNIX users from the hosts if they are marked as absent in the list